

Java Exception Handling Report

SecurityException

1. Security Exception Means

SecurityException is a Java runtime exception that occurs when a program tries to perform an operation that is not permitted by the security rules or access restrictions. In this example, the exception is deliberately generated using **throw** to demonstrate how it can be caught and handled.

2. Objective

To understand the SecurityException in Java and demonstrate how to generate, catch, and handle a SecurityException using try-catch.

3. Task Description

Write a Java program named SecurityExceptionDemo that throws a SecurityException with the message "Access denied" and catches it using a catch block. The program should display an appropriate message when the exception is handled.

4. Step-by-Step Execution

1. The program execution starts from the main() method.
2. A try block is entered.
3. The statement throw new SecurityException("Access denied") explicitly creates and throws a SecurityException.
4. Normal execution inside the try block stops immediately.
5. The matching catch (SecurityException e) block receives the exception.
6. The catch block executes System.out.println() and displays "Exception: Security access denied".
7. The program terminates normally after handling the exception.

5. Algorithm

- Step 1: Start.
- Step 2: Enter the main() method.
- Step 3: Begin the try block.
- Step 4: Throw a SecurityException with the message "Access denied".
- Step 5: Catch the SecurityException.
- Step 6: Display "Exception: Security access denied".
- Step 7: Stop.

6. UML Diagram

SecurityExceptionDemo
+ main(String[] args) try: throw SecurityException
catch (SecurityException e) System.out.println(...)

7. Program Code

```

class SecurityExceptionDemo {
    public static void main(String[] args) {
        try {
            throw new SecurityException("Access denied");
        } catch (SecurityException e) {
            System.out.println("Exception: Security access denied");
        }
    }
}

```

8. Code Explanation

class SecurityExceptionDemo — Defines a Java class named SecurityExceptionDemo.

public static void main(String[] args) — The main method is the starting point of program execution.

try — Contains code that may generate an exception.

throw new SecurityException("Access denied") — Creates a SecurityException object and explicitly throws it.

catch (SecurityException e) — Catches the SecurityException and stores it in the reference variable e.

System.out.println("Exception: Security access denied") — Displays the exception-handling message on the console.

9. Output Screenshot

Java Program Output

```
Exception: Security access denied
```

Expected console output: Exception: Security access denied